Arithmetic Sequences and Series

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Arithmetic sequences

Consider the sequence $7, 11, 15, 19, \ldots$

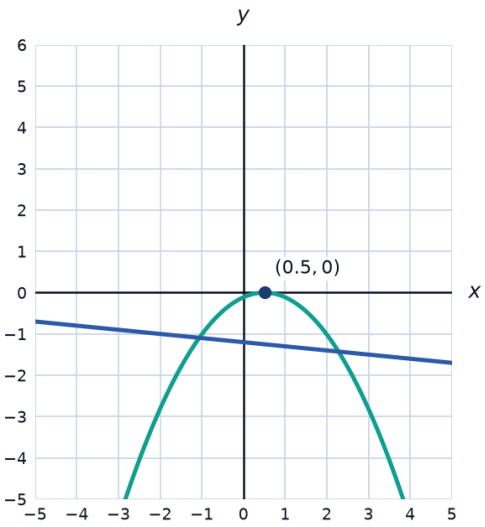
a State the common difference.

b Write a formula for the nth term.

c Find a_{20}.

2 [Foundational] • Target: Arithmetic sequences

In an arithmetic sequence, $a_3 = 10$ and $a_8 = 25$.



a Find the common difference and the first term.

b Write a formula for a_n.

3 [Foundational] • Target: Arithmetic series

Find the sum of the first 30 terms of the series $5 + 9 + 13 + \cdots$

4 [Foundational] • Target: Arithmetic series

Consider the multiples of 6 strictly between 100 and 500.

a How many are there?

b Find their sum.

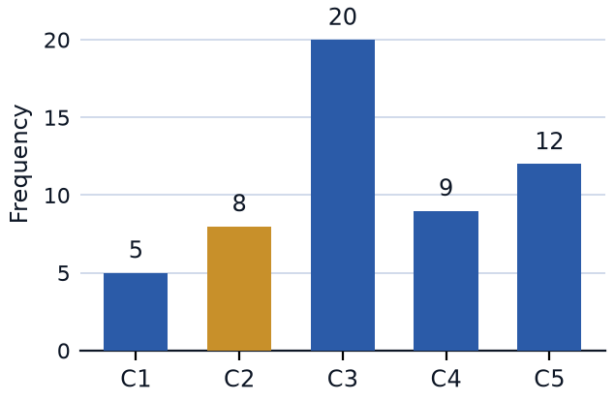
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5 [Foundational] • Target: Arithmetic series

A theatre has 18 seats in the first row, and each subsequent row has 3 more seats than the row before. How many seats are there in total in 25 rows?

6 [Foundational] • Target: Arithmetic means and sequence properties

Insert three arithmetic means between 5 and 25 (i.e., find b_1, b_2, b_3 so that $5, b_1, b_2, b_3, 25$ is arithmetic).



7 [Foundational] • Target: Arithmetic means and sequence properties

Given the arithmetic sequence with $a_1 = -8$ and $d = 3$, find the smallest n such that $a_n > 100$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Arithmetic means and sequence properties

Prove that the sum of the first n positive odd integers $(1 + 3 + 5 + \cdots)$ equals n^2, using the arithmetic series formula.

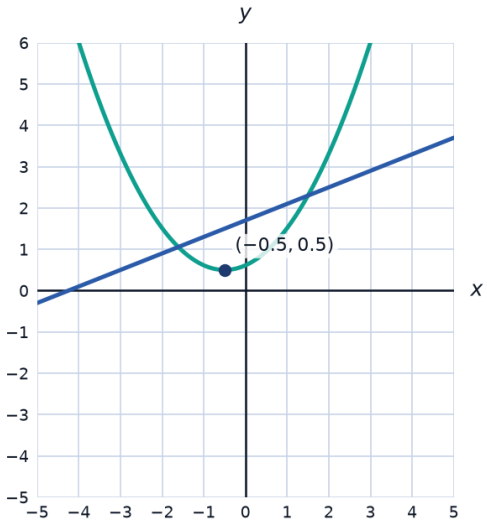
9 [Intermediate] • Target: Applications

A person saves 100 riyals in week 1, and increases savings by 20 riyals each subsequent week. Find the total saved after 20 weeks.

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10 [Intermediate] • Target: Applications

A stack of logs has 20 logs in the bottom row, 19 in the next, decreasing by 1 each row, ending with 1 log in the top row. Find the total number of logs.



11 [Intermediate] • Target: Visualizing an arithmetic sequence

The first 6 terms of the sequence $a_n = 3n - 1$ are plotted below. Use the graph to confirm the sequence is linear in n, and state the common difference from the graph.

12 [Intermediate] • Target: Arithmetic sequences in geometry

The interior angles of a pentagon form an arithmetic sequence. If the smallest angle is $80°$ and the angles sum to $540°$ (the pentagon angle sum), find the common difference.

13 [Intermediate] • Target: Finding a term given the sum

The sum of the first n terms of an arithmetic sequence with $a_1 = 3$ and $d = 4$ is $S_n = 210$. Find n.

14 [Intermediate] • Target: Arithmetic sequences with negative common difference

A sequence begins $50, 44, 38, 32, \ldots$ Find the first negative term.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

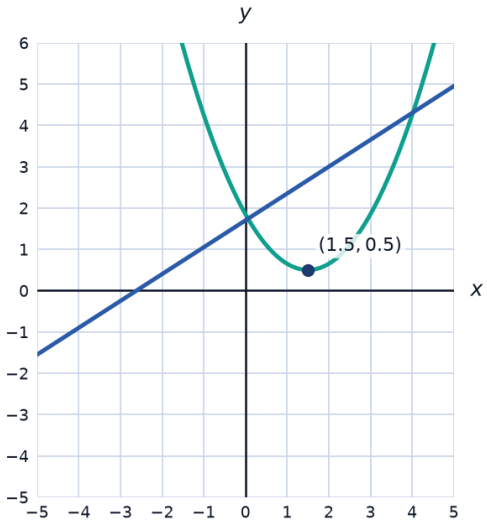
15 [Challenging] • Target: Proving an arithmetic sequence property • Cross-Lesson: 4.02 Ellipses

Prove that for any three consecutive terms a_{n-1}, a_n, a_{n+1} of an arithmetic sequence, the middle term equals the average of its neighbors: $a_n = \frac{a_{n-1} + a_{n+1}}{2}$.

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16 [Challenging] • Target: Finding a term given the sum • Cross-Lesson: 4.03 Hyperbolas

In an arithmetic sequence, the 4th term is 17 and the 10th term is 47. Find the first term a_1, the common difference d, and the sum of the first 25 terms S_{25}.



17 [Challenging] • Target: Arithmetic sequences with negative common difference • Cross-Lesson: 5.01 Vectors: Components and Operations

The sum of the first n terms of an arithmetic sequence is given by $S_n = 2n^2 + 3n$. Find the 10th term a_{10}.

18 [Challenging] • Target: Proving an arithmetic sequence property • Cross-Lesson: 5.02 The Dot Product

In an arithmetic progression, the 3rd term is 14 and the 7th term is 34. (a) Find the first term a and the common difference d. (b) Find the sum of the first 25 terms.

19 [Challenging] • Target: Arithmetic sequences • Cross-Lesson: 5.03 Parametric Equations

The sum of an arithmetic sequence is 150, with first term $a_1 = 5$ and last term $a_n = 25$. How many terms are in this sequence, and what is the common difference d?

20 [Challenging] • Target: Arithmetic series • Cross-Lesson: 6.01 Polar Coordinates and Conversions

Find the 12th term and the sum of the first 25 terms of the arithmetic sequence with $a_1 = -7$ and common difference $d = 4$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Arithmetic sequences

An arithmetic sequence has first term $a = 1$ and common difference $d = -4$. Find the sum of the first 3 terms.

a. -10

b. -7

c. -13

d. -9

22 [Foundational] • Target: Arithmetic series

An arithmetic sequence has a first term of $a_1 = 5$ and a common difference of $d = 3$. What is the value of the 6th term, a_6?

a. 20

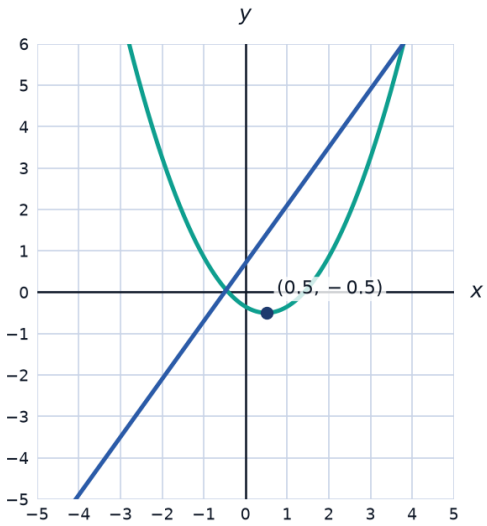
b. 23

c. 17

d. 18

23 [Foundational] • Target: Arithmetic means and sequence properties

Consider the arithmetic sequence where the first term is $a_1 = 5$ and the common difference is $d = 3$. What is the fifth term of this sequence?



a. 17

b. 14

c. 20

d. 23

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24 [Foundational] • Target: Applications

Find the sum of the first 8 terms of an arithmetic sequence with first term 5 and common difference 6.

a. 208

b. 202

c. 204

d. 206

25 [Foundational] • Target: Visualizing an arithmetic sequence

An arithmetic sequence has first term $a_1 = -7$ and common difference $d = -1$. Find a_6.

a. -2

b. -12

c. 12

d. -13

26 [Foundational] • Target: Arithmetic sequences in geometry

The first term of a geometric sequence is a. The common ratio is r. The sum of the first 3 terms is 21, and the sum of the first 6 terms is 189. If $r > 0$, find the value of a and r.

a. $a = 3, r = 2$

b. $a = 7, r = 1/2$

c. $a = 3, r = \sqrt{2}$

d. $a = 21, r = 1$

27 [Foundational] • Target: Finding a term given the sum

A geometric sequence has its first term a_1 and common ratio r. The sum of the first 3 terms is 21, and the sum of the first 6 terms is 189. Find the value of a_1 given that $req1$.

a. 1

b. 3

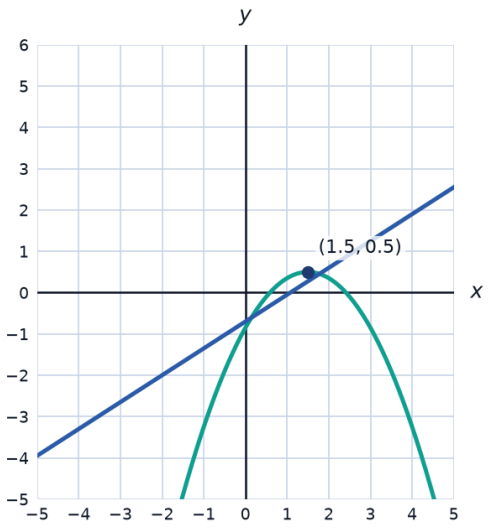
c. 7

d. 9

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28 [Foundational] • Target: Arithmetic sequences with negative common difference

An arithmetic sequence has its n-th term given by $a_n = 3n + k$, where k is a constant. If the sum of the first 5 terms of this sequence is 65, find the value of k.



a. 1

b. 2

c. 3

d. 4

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Proving an arithmetic sequence property

An arithmetic sequence has first term $a = 7$ and common difference $d = 3$. Find the 7th term.

a. 21

b. 25

c. 24

d. 26

30 [Intermediate] • Target: Arithmetic sequences

The 4th term of an arithmetic sequence is 19, and the common difference is 3. Find the first term.

a. 8

b. 9

c. 15

d. 10

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31 **[Intermediate]** • **Target: Arithmetic series**

An arithmetic sequence has first term 10 and last term 40 with 7 terms. Find the sum.

a. 140

b. 175

c. 280

d. 350

32 **[Intermediate]** • **Target: Arithmetic means and sequence properties**

What is the common difference of the arithmetic sequence $0, -2, -4, -6, \ldots$?

a. 0

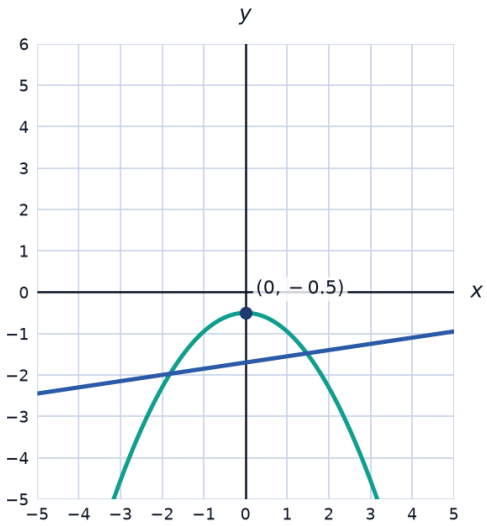
b. -2

c. -4

d. -3

33 **[Intermediate]** • **Target: Applications**

An arithmetic sequence has first term 6 and common difference 5. Find n such that $S_n = 1403$.



a. 19

b. 25

c. 27

d. 23

34 **[Intermediate]** • **Target: Visualizing an arithmetic sequence**

A geometric sequence has first term $a = 1$ and common ratio $r = 2$. Find the sum of the first 5 terms.

a. 31

b. -31

c. 32

d. 16

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35 [Intermediate] • Target: Arithmetic sequences in geometry

In an arithmetic sequence, $a_4 = -1$ and $a_8 = -9$. Find the common difference d.

a. -2

b. -3

c. -6

d. 6

36 [Intermediate] • Target: Finding a term given the sum

The 12th term of an arithmetic sequence is 58, and the first term is 14. Find the common difference.

a. 4

b. 1

c. 6

d. -1

37 [Intermediate] • Target: Arithmetic sequences with negative common difference

Consider an infinite geometric series with first term a and common ratio r, where $|r| < 1$: $a + ar + ar^2 + ar^3 + \ldots = 12$ If each term in the series is squared, the sum of the resulting infinite geometric series is: $a^2 + a^2r^2 + a^2r^4 + a^2r^6 + \ldots = 72$ What is the value of the common ratio, r?

a. $\frac{1}{3}$

b. $\frac{1}{2}$

c. $\frac{2}{3}$

d. $\frac{1}{4}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Proving an arithmetic sequence property • Cross-Lesson: 5.02 The Dot Product

In a geometric sequence of positive real numbers, the sum of the first two terms is 18, and the sum of the first four terms is 90. What is the fifth term of the sequence?

a. 48

b. 72

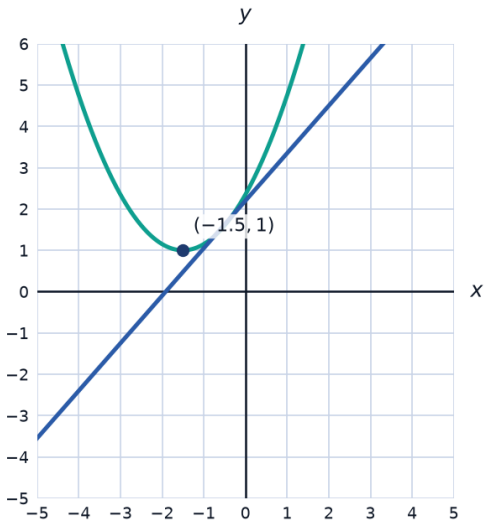
c. 96

d. 192

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39 [Challenging] • Target: Arithmetic sequences • Cross-Lesson: 5.03 Parametric Equations

An arithmetic sequence begins with the terms: $8, 13, 18, 23, \ldots$ What is the 15th term of the sequence?



a. 78

b. 83

c. 73

d. 117

40 [Challenging] • Target: Arithmetic series • Cross-Lesson: 6.01 Polar Coordinates and Conversions

A geometric sequence has its first term $a_1 = 3$ and a common ratio $r = 2$. What is the third term, a_3, of this sequence?

a. 6

b. 9

c. 12

d. 15

41 [Challenging] • Target: Arithmetic means and sequence properties • Cross-Lesson: 6.02 Polar Equations and Graphs

Find the first term and common difference of the AP whose pth term is $3p + 2$.

a. $a_1 = 2, d = 3$

b. $a_1 = 5, d = 3$

c. $a_1 = 5, d = 2$

d. $a_1 = 3, d = 2$

42 [Challenging] • Target: Applications • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

The 5th term of the arithmetic sequence $2, 5, 8, 11, \ldots$ is:

a. 15

b. 13

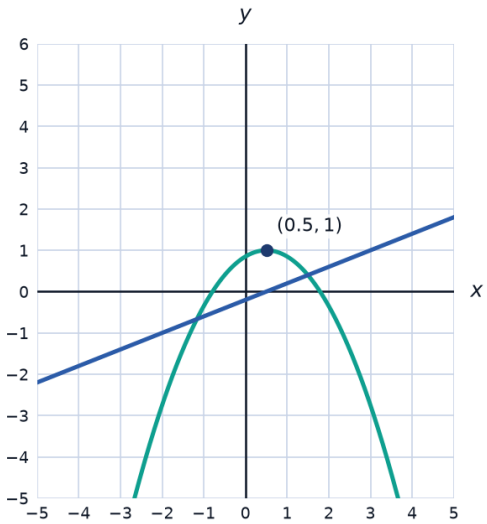
c. 12

d. 14

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43 [Challenging] • Target: Visualizing an arithmetic sequence • Cross-Lesson: 7.02 Logarithms and Their Properties

An arithmetic sequence has $a_3 = 0$ and $a_5 = -4$. Find S_{12}, the sum of the first 12 terms.



a. -92

b. -84

c. -85

d. -79

44 [Challenging] • Target: Arithmetic sequences in geometry • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

An arithmetic sequence has $a_1 = 3$ and $d = 6$. If the sum of the first n terms is 300, find n.

a. 11

b. 9

c. 5

d. 10

45 [Challenging] • Target: Finding a term given the sum • Cross-Lesson: 7.04 Applications: Growth and Decay

Find the 15th term of an arithmetic sequence where $a_3 = 11$ and $a_7 = 23$.

a. 39

b. 51

c. 47

d. 43

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